Voluntary buffers under finite common coverage

The same reserve rate applies to banks with different liquidity exposure and interbank access.

1. Each bank chooses how much liquidity to keep before the shock.
2. Interbank trade redistributes available surplus.
3. A finite common capacity covers part of the remaining shortfall.

If protection is available later, how much self-insurance is worth holding now?

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github.com/carlosgomezp28/ai-project

The current post-Marco benchmark

One currency, interpreted as USD; two bank types.

$$\begin{array}{ccccc} t=0 & \longrightarrow & t=1 & \longrightarrow & t=2 \\ \text{Choose buffer } u_k & \longrightarrow & \begin{array}{c}\text{Shock} \to \text{interbank trade} \\ \to \text{common coverage}\end{array} & \longrightarrow & \text{Settlement} \end{array}$$

- Required reserves are **not individually usable (NU)**.
- Allocation and interbank priority are **prescribed**; i is given.
- Common coverage is free in the benchmark and has finite capacity.

Joint thesis architecture: Carlos Gómez Puicán and Fabián Méndez Suárez, under Marco Ortiz, incorporating his benchmark note.

Individual course contribution: buffer optimization, verification and simulations. Open choices are not treated as advisor-approved.

Thesis: heterogeneous incidence
How does relative common coverage differ across banks with different exposure, access and self-insurance?

$$I_k = \frac{C_k/d_k}{\bar{C}/D} \qquad \text{relative coverage, not a net transfer}$$

Individual AI-course question
How does a bank optimally choose voluntary liquidity buffers when interbank access and finite common coverage shape the marginal value of liquidity?

Narrowing the question: solve and check the individual best response before claiming an aggregate incidence result.

Signed liquidity flow and market accounting

$$\omega_k = \beta_k A + \varepsilon_k, \qquad s_k = d_k(\omega_k + u_k)$$
$$D_k = \mathbb{E}[(-s_k)^+ \mid A], \qquad X_k = \mathbb{E}[(s_k)^+ \mid A]$$

Negative s_k: shortfall. Positive s_k: surplus. D_k, X_k are **amounts**; u_k is a ratio to deposits.

$$\psi_k = \tau_k \min\{1, P/Q\}, \qquad \ell_k = \tau_k \min\{1, Q/P\}$$

Access $\tau_k \in [0,1]$; $P = \sum_k \mu_k \tau_k X_k$, $Q = \sum_k \mu_k \tau_k D_k$.

$$\underbrace{\sum_k \mu_k \psi_k D_k}_{\text{borrowed}} = \underbrace{\sum_k \mu_k \ell_k X_k}_{\text{lent}} = \min\{P, Q\}$$

Trade redistributes liquidity; it does not create it.

The bank's optimization problem

Expected private cost per deposit, $J_k(u_k)$

$$\min_{u_k \in \mathcal{U}_k} \left\{ r u_k + \mathbb{E}_A \left[i\psi_k \frac{D_k}{d_k} - i\ell_k \frac{X_k}{d_k} + \kappa(1-\varphi)(1-\psi_k)\frac{D_k}{d_k} \right] \right\}.$$

Carrying cost	Liquidity held before knowing the shock
Interbank financing	Pay on the covered deficit
Lending income	Earn on the surplus actually lent
Uncovered-shortfall cost	Pay for the final residual

The atomistic bank takes the **state-contingent schedules** $\psi_k(A), \ell_k(A), \varphi(A)$ and prices as given.

It knows the rules, not realized A or ε_k.

Current benchmark equation (7); $\kappa > i \geq 0$. All objective terms are per unit of deposits.

Proposition 1: the marginal value of a buffer

$$\pi_k^- = \mathbb{P}(\omega_k + u_k < 0 \mid A), \quad \pi_k^+ = \mathbb{P}(\omega_k + u_k > 0 \mid A)$$

$$\frac{\partial(D_k/d_k)}{\partial u_k} = -\pi_k^-, \qquad \frac{\partial(X_k/d_k)}{\partial u_k} = +\pi_k^+.$$

$$\boxed{r = \mathbb{E}_A\big[\big(i\psi_k + \kappa(1-\varphi)(1-\psi_k)\big)\pi_k^- + i\ell_k\pi_k^+\big]}$$

Marginal carrying cost = avoided deficit cost + additional lending opportunity.

Conditions: integrable signed shocks; atomless conditional law; an interior choice; fixed schedules for this individual best response.

This is a necessary FOC; sufficiency needs additional conditions.

Why the FOC is not automatically sufficient

$$a_k = i\psi_k + \kappa(1-\varphi)(1-\psi_k), \qquad b_k = i\ell_k$$

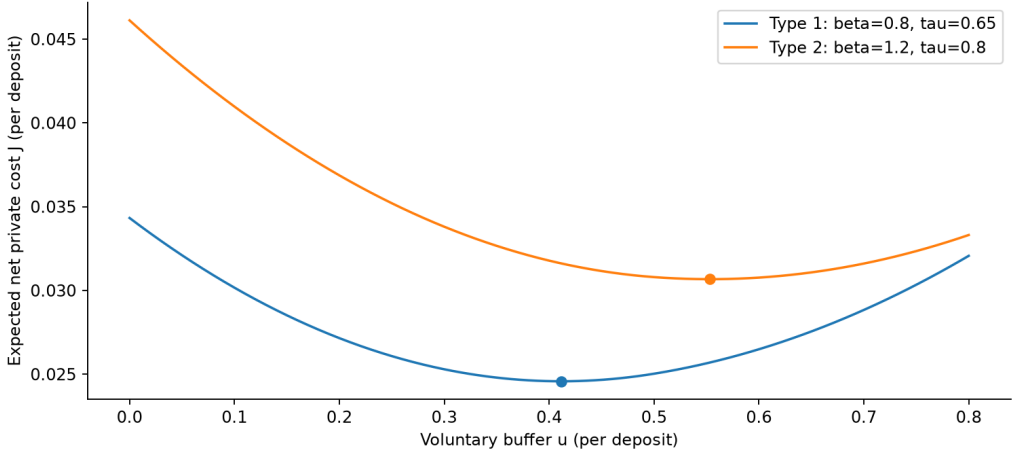
$$J_k''(u) = \mathbb{E}_A[(a_k - b_k) f_{\omega_k|A}(-u)].$$

- Avoided deficit costs add curvature.
- **Lending income subtracts curvature.**
- A sufficient global condition is $a_k(A) \geq b_k(A)$ state by state.

Otherwise: compare stationary points and corners.
$\kappa > i$ alone does not establish convexity.

Density regularity is needed for J''. With global convexity, an interior stationary point with positive local curvature is the unique global minimum.

Simulation: cost curves and optimal buffers



Type 1: $u_1^* = 0.411523$ **Type 2:** $u_2^* = 0.553381$ (unique, interior)

Constructed, not Peruvian estimates; large illustrative buffers. Vertical axis truncated.

Proposition 2: common capacity and self-insurance

$$\varphi(A) = \min\{1, F/\Delta(A)\}, \qquad F = \nu_F \rho D + R$$

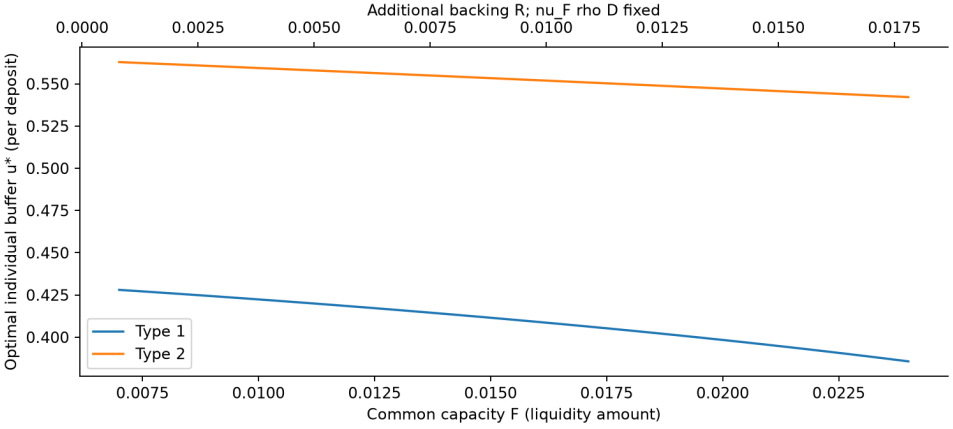
$$\boxed{\frac{du_k^*}{dF} = -\frac{\kappa\, \mathbb{E}_A\left[1_{\{0<F<\Delta(A)\}} \frac{(1-\psi_k)\pi_k^-}{\Delta(A)}\right]}{J_k''(u_k^*)}}$$

- Regular interior optimum, positive curvature; no coverage kink.
- Strict sign needs a rationed state with residual deficit exposure.
- $\psi_k, \ell_k, \Delta, P, Q$, prices, law and domain remain fixed.
- Local density/IFT regularity; off support knots in the uniform code.

More capacity reduces the private value of self-insurance.

Conditional best response; not general equilibrium. With ν_F, ρ, D fixed, $du^*/dR = du^*/dF$.

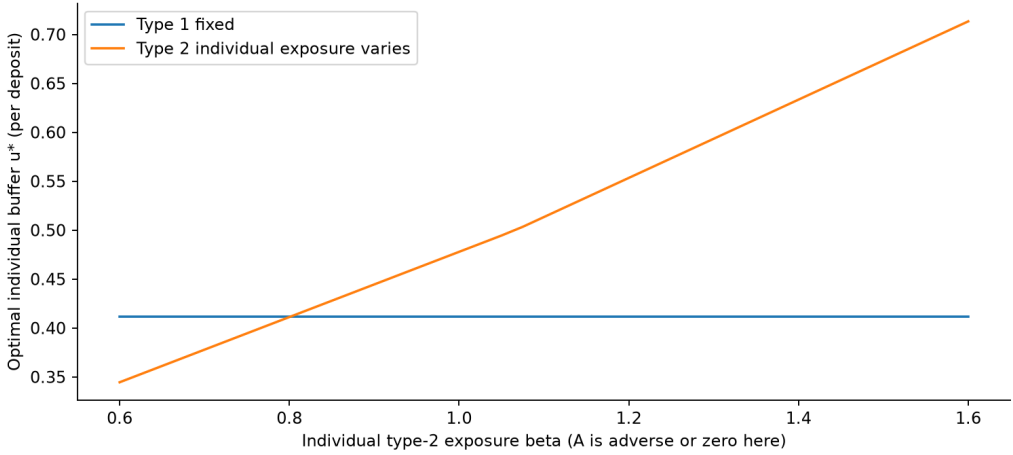
Simulation: the conditional capacity response



At $F = 0.015$: $du_1^*/dF = -2.377540$; $du_2^*/dF = -1.214916$.

Conditional best response, not general equilibrium. With fixed buffers and fully rationed prorrata, absolute coverage rises; I_k is invariant. Constructed inputs; vertical axis truncated.

Simulation: exposure with access held fixed



$\beta_2 : 0.6 \to 1.6$, $u_2^* : 0.345 \to 0.713$; access/schedules fixed; Type 1 unchanged.

Constructed grid, $A \in \{-0.4, 0\}$; no universal risk theorem. Vertical axis truncated.

What Lean verifies, and the human unit check

Explicit finite-uniform building block

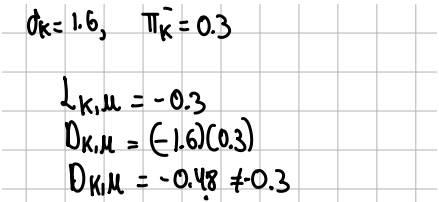
$$\widehat{L}'(t) = -\sum_j w_j \Pi_{\sigma_j}(t - m_j).$$

Positive widths; every component off its support knots.
`finiteLoss_derivative`: type and proof excerpts

```
HasDerivAt (finiteLoss w sigma mean)
  (-finiteTail w sigma mean t) t := by

convert hd using 1
simp [finiteTail, Finset.sum_neg_distrib]
```

Student's handwritten unit check



Authentic excerpt: manuscript 02, p. 2. The amount derivative carries d_k; the ratio derivative does not.

Existing Lean proofs verify off-knot derivatives of the explicit finite-uniform positive-part building block. The post-Marco probability interpretation, buffer FOC, capacity response, and equilibrium are **not newly Lean-verified**.

Open mechanisms and questions for discussion

Still open

- Voluntary participation versus prescribed interbank allocation.
- Aggregate consistency: no population fixed point solved here.
- Shock laws and buffer domain are illustrative course choices.

Feedback

1. Is prescribed interbank allocation an acceptable benchmark for the course/thesis transition?
2. Is the conditional best response informative before solving aggregate consistency?
3. What comes next: participation incentives, shock/domain discipline, or the aggregate fixed point?

Backup material

Accounting, constructed assumptions, proof scope and model history

All moments and schedules below are conditional on A.

$$\omega_k = \beta_k A + \varepsilon_k, \quad s_k = d_k(\omega_k + u_k) \tag{1}$$

$$D_k = \mathbb{E}[(-s_k)^+ \mid A], \quad X_k = \mathbb{E}[(s_k)^+ \mid A] \tag{2}$$

$$\psi_k = \tau_k \min\{1, P/Q\}, \quad \ell_k = \tau_k \min\{1, Q/P\} \tag{3}$$

$$\sum_k \mu_k \psi_k D_k = \sum_k \mu_k \ell_k X_k = \min\{P, Q\}. \tag{4}$$

$$P = \sum_k \mu_k \tau_k X_k, \quad Q = \sum_k \mu_k \tau_k D_k, \quad D = \sum_k \mu_k d_k.$$

If $P = 0$ or $Q = 0$, transacted volume is zero (implemented fractions $\psi_k = \ell_k = 0$).

Current thesis candidate, analytical framework (1)–(4). Prescribed quantities are not voluntary participation conditions.

$$F = \nu_F \rho D + R, \qquad \Delta = \sum_k \mu_k (1 - \psi_k) D_k,$$
$$\varphi = \min\{1, F/\Delta\}, \quad C_k = \mathbb{E}_A[\varphi(1 - \psi_k) D_k]. \tag{5}$$

If $\Delta = 0$, set $\varphi = 0$; common coverage has price $p_F = 0$.

$$\bar{C} = \sum_k \mu_k C_k, \qquad I_k = \frac{C_k/d_k}{\bar{C}/D}, \quad \bar{C} > 0. \tag{6}$$

$$\min_{u_k \in \mathcal{U}_k} \left\{ r u_k + \mathbb{E}_A \left[i \psi_k \frac{D_k}{d_k} - i \ell_k \frac{X_k}{d_k} + \kappa (1 - \varphi)(1 - \psi_k) \frac{D_k}{d_k} \right] \right\}. \tag{7}$$

$\nu_F \in [0, 1]$, $R \geq 0$, $\kappa > i$. NU reserves are not also individually usable liquidity. I_k is undefined if $\bar{C} = 0$.

Current thesis candidate (5)–(7). Net transfers require property, repayment and loss rules not supplied here.

B3 · Constructed course parameters

	Type 1	Type 2
Mass μ_k	0.6	0.4
Deposits d_k	1	1.6
Exposure β_k	0.8	1.2
Access τ_k	0.65	0.80
Reference u_k	0.10	0.10
Course upper bound	0.8	0.8

Common primitives	Value
Carrying cost r	0.045
Interbank rate i	0.015
Uncovered cost κ	0.25
Reserve rate ρ	0.10
Capacity share ν_F	0.05
Backing R	0.0088
Baseline F	0.015

$A = (-0.4, 0)$ with probabilities $(0.4, 0.6)$; $\varepsilon_k \mid A \sim U[-0.5, 0.5]$. Aggregate deposits $D = 1.24$. Finite states, uniforms and $[0, 0.8]$ are **course assumptions**.

Schedules come from reference buffers, not the displayed best responses.

Source: code/output/post_marco/parameters.json. All parameters constructed; no Peruvian estimates or advisor approval implied.

State	P	Q	Borrowed = lent	Fund used
Adverse	0.0189744	0.2993344	0.0189744	0.015
Normal	0.1623600	0.0721600	0.0721600	0.015

$$\sum_k \mu_k \psi_k D_k = \min\{P, Q\} = \sum_k \mu_k \ell_k X_k,$$

$$\sum_k \mu_k \varphi(1-\psi_k) D_k = \min\{F, \Delta\} \le F.$$

Reference Δ: adverse 0.3843536; normal 0.02704.

Checks also cover $P < Q$, $P > Q$, $P = Q$, zero sides/access, $\Delta = 0$, zero capacity and saturated coverage.

These are reference-population accounting checks.

They do not make frozen-schedule best responses jointly consistent.

Sources: parameters.json, reference_market_accounting.csv, and verify.py: 99 checks across the three course models.

$$X_k/d_k = D_k/d_k + \mathbb{E}[\omega_k \mid A] + u$$

$$J(u) = (r - \mathbb{E}[b_k])u + \mathbb{E}[(a_k - b_k)L_k] - \mathbb{E}[b_k \mathbb{E}[\omega_k \mid A]].$$

Thus $a_k \geq b_k$ gives global convexity. Otherwise this proof fails.

Counterexample to automatic convexity:
Fully covered deficits give $\varphi = 1$ and $a_k - b_k = i(\psi_k - \ell_k)$. In the adverse reference state, $\psi_1 < \ell_1$. With $F = 10$, $J_1''(0.65) \simeq -0.003653$.

Under convexity on $[0, \bar{u}]$:

$$J'(0) \geq 0 \Rightarrow 0 \text{ minimizes}, \qquad J'(\bar{u}) \leq 0 \Rightarrow \bar{u} \text{ minimizes}.$$

Equalities can permit ties. The code compares every quadratic-region endpoint/stationary point, including nonconvex and flat cases.

Source: verify_post_marco.py; finite-uniform course specialization.

$$\bar{C} = \sum_k \mu_k C_k, \quad I_k = \frac{C_k/d_k}{\bar{C}/D}, \quad \sum_k \frac{\mu_k d_k}{D} I_k = 1.$$

When every state with residual deficit remains rationed and buffers and allocation stay fixed,

$$C_k = F\,\mathbb{E}_A\left[\frac{(1-\psi_k)D_k}{\Delta}\right].$$

F scales absolute coverage and cancels from relative incidence.

In this constructed sweep: $I_1 = 1.091124$, $I_2 = 0.914571$; neither changes with F.

These are **fixed-reference-buffer** values. Incidence with jointly reoptimized buffers needs aggregate consistency.

If $\bar{C} = 0$, I_k is undefined. Coverage shares do not identify net transfers, subsidy or welfare.

Source: fixed_buffer_incidence.csv. Proportional scaling does not imply directional heterogeneous incidence.

B7 · Exact conditions for the capacity result

1. Fixed deposits, domain, shock law, r, i, κ; fixed $\psi_k, \ell_k, \Delta, P, Q$. Only φ changes with F.
2. Regular differentiable interior optimum branch, $J''(u^*) > 0$; no competing global-minimum switch.
3. Valid derivative/expectation interchange. Finite-state uniforms: positive widths and no relevant support knot.
4. $F > 0$, no relevant $F = \Delta(A)$ kink. If $\Delta = 0$, use the stated zero-coverage convention.
5. Strict negativity requires positive-probability states with $F < \Delta$, $\psi_k < 1$, and $\pi_k^- > 0$. Otherwise only a weak sign is justified.

The implicit-function argument gives a **local** response. Global best-response language needs global sufficiency/stability.

Holding ν_F, ρ, D fixed gives $du^*/dR = du^*/dF$. It does not give a reserve-rate, GE or participation theorem.

POST_MARCO_AUDIT.md, section 4. Corners, ties, saturation and one-sided boundary questions are separate.

	October	November 4	Current post-Marco
Core	Two currencies; public dollar backing	Direct support; asset-linked eligibility	One USD currency; prescribed interbank allocation
Buffer benefit	Avoid deficit costs	Avoid financing and deferral costs	Avoid deficit costs AND earn lending income
φ meaning	Common/public coverage	Reserve usability	Common coverage fraction
Comparative static	Conditional backing effect	Conditional reserve-burden effect	Conditional capacity effect

Historical work is preserved; its equations are not silently reused.

Existing Lean package remains pinned to paper `cf37d3f`. Current economic source: accepted post-Marco computational checkpoint.

B9 · Existing Lean proof and the new interpretation

`Gomez26EligibilityBuffers.ell_derivative` (`BufferCore.lean`), unchanged:

```
theorem ell_derivative {sigma z : ℝ} (hs : 0 < sigma)
    (hnl : z ≠ -sigma) (hnu : z ≠ sigma) :
    HasDerivAt (ell sigma) (-piTail sigma z) z := by
  by_cases hl : z ≤ -sigma
  · simpa [piTail, hl] using ell_derivative_below
      (lt_of_le_of_ne hl hnl)
  · by_cases hu : z < sigma
    · simp only [piTail, if_neg hl, if_pos hu]
      convert ell_derivative_inside hs
        (lt_of_not_ge hl) hu using 1; ring
    · simpa [piTail, hl, hu] using
        ell_derivative_above hs
          (lt_of_le_of_ne (le_of_not_gt hu) hnu.symm)
```

New deficit input: $z = u + \beta A$ (equivalently finite-loss means $m_j = -\beta A_j$).
Reflecting the symmetric uniform shock gives the probability interpretation; that bridge and the new economic propositions are not already Lean proofs.